GECO: a calibrated GEFS intensity ensemble from public real-time data

GEFS Ensemble, Consensus-Oriented · architecture summary · 29 October 2026 · North Atlantic and East Pacific

What it is

GECO turns the 30 GEFS member tracks that NHC publishes in real time into an intensity ensemble that is centred correctly, has a realistic and storm-specific spread, and keeps each member's physically consistent scenario (when it intensifies, peaks and weakens). It runs at every synoptic cycle (00, 06, 12, 18 UTC), uses only NHC's public files, and is ready within the early-cycle window, like NHC's own consensus aids. Each output member is

$$\text{member}_i(\tau) = \text{HCCA}(\tau) + k \cdot \hat{\sigma}(\tau) \cdot z_i(\tau),$$

where HCCA is NHC's corrected consensus forecast (the *centre*), $\hat{\sigma}$ is a predicted forecast error (the *size* of the spread) and z_i is the member's normalised scenario from the GEFS (the *shape*). Each of the three comes from the source that does it best.

Inputs (all available by about t_0 + 3 h)

Input	Source and use
GEFS member tracks (AP01–30)	NHC's public a-decks; the newest run at least 6 h old (the 00 UTC run serves the 06 UTC cycle), leads shifted by the run's age. Provides the scenario shapes z_i.
HCCA	NHC's corrected consensus, in the same public a-decks at every cycle. Provides the centre. It draws on ECMWF guidance that NHC does not publish in real time.
Storm state (CARQ) and early aids	NHC's operational analysis at t_0 and the twelve interpolated aids (GFS, GEFS mean, UKMET, Canadian, Navy, HWRF/HMON/HAFS, COAMPS-TC, DSHP, LGEM). Feed the spread model.
Our consensus correction (TCCA)	A LightGBM correction of the early-aid mean (no ECMWF, no SHIPS). Its output and quantile width feed the spread model, and it is the fallback centre when HCCA is missing (0.3% of 2025–26 forecasts).

Processing steps

1. Scenario shapes. For member i at lead τ (every 12 h to 168 h), take its departure from the ensemble mean and re-anchor it so that every member starts together at t_0: $d_i(\tau) = [v_i(\tau) - \overline{v}(\tau)] - [v_i(0) - \overline{v}(0)]$. Smooth d_i along lead with a 6-h Gaussian kernel (this gives the members the step-to-step statistics of real best tracks) and divide by the forecast's own spread of smoothed departures, itself smoothed along lead (18 h), to get z_i with unit spread. The raw GEFS intensity level, about 25 kt too weak, is discarded; only the shape is kept.

2. Spread size. A LightGBM regression predicts $\hat{\sigma}$, the expected absolute error of a consensus forecast for this storm, cycle and lead. Features: the disagreement among the early aids, the spread of the GEFS member tracks, the storm's intensity, recent trend, position and distance to land, the lead time, and our consensus correction's quantile width and predicted 24-h change. It is trained on out-of-fold forecast errors from 2018–2024 (leave-one-season-out), so it learns from seven seasons of realistic errors. $\hat{\sigma}$ is smoothed along lead (18 h) and floored at 2 kt.

3. Centre. HCCA as issued at its 12-hourly leads, with no further smoothing. Gaps between HCCA leads are interpolated; past HCCA's last lead the last HCCA-minus-TCCA offset is carried forward.

4. Combine. $\text{member}_i = \text{HCCA} + k \cdot \hat{\sigma} \cdot z_i$, clipped to 10–200 kt. The factor $k = 1.2$ (constant with lead) converts a mean absolute error into an ensemble spread; it is the value that minimises the fair CRPS on out-of-fold 2018–2024 cases. The full fitted spread is used. Because the z_i average to zero, the ensemble mean equals HCCA: GECO adds a distribution, not a new mean.

Fitted component	Training data	Validation
Consensus correction (feeds $\hat{\sigma}$)	2018–2024, leave-one-season-out	2025–26 held out
Spread model $\hat{\sigma}$	out-of-fold errors 2018–2024, leave-one-season-out	2025–26 held out
Spread factor k	out-of-fold CRPS 2018–2024	refit without each season: 1.2–1.3
Smoothing widths (6 h, 18 h)	chosen on step-to-step statistics vs the best track	—

How well it works

Verified against NHC's best track (2026: working best track), leads 12–120 h, all four cycles, identical cases for every system, 95% storm-block bootstrap intervals. For 2018–2024 every fitted part is out of sample for the season scored. The

fair probabilistic yardstick is "dressed HCCA": HCCA as a normal distribution whose width at each lead is HCCA's own error in the other seasons.

Fair CRPS (kt; lower is better)	**2018–24**	**2025–26**	**All 2018–26**
Raw GEFS members (offset to t_0)	12.69	14.61	13.10
Dressed OFCL	7.49	7.68	7.53
Dressed HCCA	7.33	7.58	7.39
GECO	**6.97**	**7.26**	**7.04**
GECO − dressed HCCA	−0.36 [−0.43, −0.29]	−0.32 [−0.43, −0.20]	−0.36 [−0.41, −0.29]
GECO 10–90% coverage (ideal 80%)	77%	76%	76%

336 storms, 27,623 cases. GECO beats dressed HCCA in each of the nine seasons separately, each significantly, across the GEFS upgrade from 20 to 30 members (2020) and the HWRF-to-HAFS switch (2023).

Against FNV3 (Google DeepMind), 2025–26. On 6,405 identical cases (61 storms), GECO's CRPS is 7.6 kt against 8.4 kt for FNV3′s 50 members taken at the same early timing: −0.78 kt [−1.60, −0.01], significant. GECO is better or level at every lead to 120 h; its mean bias is −1.2 kt against FNV3′s −6.0 kt; its spread is 0.87 of its error against 0.69 for FNV3, which is over-confident.

Physical consistency. GECO members change by 9.0 kt per 12 h on average (best track 8.6 kt), zig-zag less often than the best track (0.025 vs 0.034) and exceed 30 kt in 12 h about as often (2.1% vs 1.9%).

Real-time operation and limits

At each cycle: read NHC's a-decks, take the newest GEFS run at least 6 h old, compute z_i, run the spread model, read HCCA, combine. It is a few seconds of computation, with no gridded data. Arrival times of HCCA and the GEFS member tracks relative to the forecast deadline have not yet been measured.

Limits: the mean is exactly HCCA, so GECO inherits HCCA's errors, including its low bias during rapid intensification. The spread is symmetric about the centre, so rapid intensification still falls above the 90th percentile more often than nominal. The spread model has not been retrained on HCCA's errors (it predicts our consensus correction's errors, and k absorbs the difference).

Code: `src/tcic/ensemble/geco.py`, `scripts/ens_geco.py`; nine-season evaluation `scripts/ens_multiseason.py`, `docs/ensemble_multiseason.md`; FNV3 comparison `scripts/fnv3_compare.py`.